

# References

ABC News Nightline (1999). *Deep Dive*.

Abernethy, C. N. (1993). Expanding jurisdictions and other facets of human-machine interface IT standards. *Standard View*, 1(1), 9–21. <https://doi.org/10.1145/174683.174685>.

Acohido, B. (1999). Did similar switches confuse pilots?—Controls' proximity another aspect of crash probe. In: *Seattle Times Investigative Reporter*. November 18, Retrieved from <http://community.seattletimes.nwsourc.com/archive/?date=19991118&slug=2996058>.

Adams, D. (1990). *The long dark tea-time of the soul* (1st ed.). New York, NY: Pocket Books.

Adler, C. (2011). Ideas are overrated: startup guru Eric Ries' radical new theory. *Wired Magazine*, 19(09), 34.

Alben, L., Faris, J., & Saddler, H. (1994). Making it Macintosh: designing the message when the message is design. *interactions*, 1(1), 11–20. <https://doi.org/10.1145/174800.174802>.

Albers, J. (1974). *Interaction of color*. New Haven, CN: Yale University Press.

Alvarez, H. (2014). *A guide to color, UX, and conversion rates*. Retrieved from <https://www.usertesting.com/blog/2014/12/02/color-ux-conversion-rates/>.

Ann, E. (2009). What's design got to do with the world financial crisis? *interactions*, 16(3), 20–27.

Antle, A. N. (2009). Embodied child computer interaction: why embodiment matters. *interactions*, 16(2), 27–30.

Apple Computer Inc. (1993). *Making it Macintosh: The Macintosh human interface guidelines companion*. Reading, MA: Addison-Wesley.

Arnowitz, J. (2013). Taking the fast RIDE: designing while being agile. *interactions*, 20(4), 76–79. <https://doi.org/10.1145/2486227.2486243>.

Bangor, A., Kortum, P. T., & Miller, J. T. (2008). An empirical evaluation of the system usability scale. *International Journal of Human Computer Interaction*, 24(6), 574–594.

Barnard, P. (1993). The contributions of applied cognitive psychology to the study of human-computer interaction. In: R. M. Baecker, J. Grudin, B. Buxton, & S. Greenberg (Eds.), *Readings in human computer interaction: Toward the year 2000* (pp. 640–658). San Francisco, CA: Morgan Kaufmann.

Baskinger, M. (2008). Pencils before pixels: a primer in hand-generated sketching. *interactions*, 15(2), 28–36.

Baskinger, M., & Gross, M. (2010). Tangible interaction = Form + computing. *interactions*, 17(1), 6–11.

Bastien, J. M. C., & Scapin, D. L. (1995). Evaluating a user interface with ergonomic criteria. *International Journal of Human Computer Interaction*, 7(2), 105–121.

Baty, S. (2010). Solving complex problems through design. *interactions*, 17(5), 70–73.

Beale, R. (2007). Slanty design. *Communications of the ACM*, 50(1), 21–24. <https://doi.org/10.1145/1188913.1188934>.

Beck, K. (1999). Embracing change with extreme programming. *IEEE Computer*, 32(10), 70–77.

- Beck, K. (2000). *Extreme programming explained: Embrace change*. Boston, MA: Addison-Wesley.
- Beck, K., & Andres, C. (2004). *Extreme programming explained: Embrace change, 2nd edition (the XP series)* (2nd ed.). Boston, MA: Addison-Wesley.
- Becker, K. (2004). Log on, tune in, drop down: (and click “go” too!). *interactions*, 11(5), 30–35. <https://doi.org/10.1145/1015530.1015543>.
- Bell, T. E., & Thayer, T. A. (1976). Software requirements: are they really a problem? In: *Proceedings of the 2nd international conference on software engineering, San Francisco, California, USA*.
- Benington, H. D. (1956). United States, navy mathematical computing advisory panel. In: *Symposium on advanced programming methods for digital computers*. Washington, DC: Office of Naval Research, Dept. of the Navy.
- Benington, H. D. (1983). Production of large computer programs. *IEEE Annals of the History of Computing*, 5(4), 350–361. <https://doi.org/10.1109/mahc.1983.10102>.
- Bennett, J. L. (1984). Managing to meet usability requirements: establishing and meeting software development goals. In: J. Bennett, D. Case, J. Sandelin, & M. Smith (Eds.), *Visual display terminals* (pp. 161–184). Englewood Cliffs, NJ: Prentice-Hall.
- Bera, P. (2016). How colors in business dashboards affect users’ decision making. *Communications of the ACM*, 59(4), 50–57.
- Berry, R. E. (1988). Common user access—a consistent and usable human-computer interface for the SAA environments. *IBM Systems Journal*, 27(3), 281–300. <https://doi.org/10.1147/sj.273.0281>.
- Beyer, H., & Holtzblatt, K. (1997). *Contextual design: A customer-centered approach to systems designs* (1st ed.). San Francisco, CA: Morgan Kaufmann.
- Beyer, H., & Holtzblatt, K. (1998). *Contextual design: Defining customer-centered systems* (1st ed.). San Francisco, CA: Morgan-Kaufman.
- Beyer, H., Holtzblatt, K., & Baker, L. (2004). An agile customer-centered method: rapid contextual design. In: *Extreme programming and agile methods (LNCS 3134)* (pp. 50–59). Calgary, Canada: Springer Berlin/Heidelberg.
- Bias, R. G., & Mayhew, D. J. (2005). *Cost-justifying usability: An update for the internet age* (2nd ed.). San Francisco, CA: Morgan Kaufmann.
- Billingsley, P. A. (1993). Reflections on ISO 9241: software usability may be more than the sum of its parts. *Standard View*, 1(1), 22–25. <https://doi.org/10.1145/174683.174686>.
- Bittner, K., & Spence, I. (2003). *Use case modeling*. Boston, MA: Addison-Wesley.
- Bjerknes, G., Ehn, P., & Kyng, M. (Eds.), (1987). *Computers and democracy: A Scandinavian challenge*. Aldershot, UK: Avebury.
- Blythin, S., Rouncefield, M., & Hughes, J. A. (1997). Never mind the ethno’ stuff, what does all this mean and what do we do now: ethnography in the commercial world. *interactions*, 4(3), 38–47. <https://doi.org/10.1145/255392.255400>.
- Bødker, S. (1991). *Through the interface: A human activity approach to user interface design*. Hillsdale, NJ: Lawrence Erlbaum.
- Bødker, S. (2015). Third-wave HCI, 10 years later—participation and sharing. *interactions*, 22(5), 24–31. <https://doi.org/10.1145/2804405>.
- Bødker, S., & Buur, J. (2002). The design collaboratorium—a place for usability design. *ACM Transactions on Computer-Human Interaction*, 9(2), 152–169.
- Bødker, S., Ehn, P., Kammersgaard, J., Kyng, M., & Sundblad, Y. (1987). A utopian experience. In: G. Bjerknes, P. Ehn, & M. Kyng (Eds.), *Computers and Democracy—A Scandinavian Challenge* (pp. 251–278). Aldershot, UK: Avebury.
- Boehm, B. W. (1988). A spiral model of software development and enhancement. *IEEE Computer*, 21(5), 61–72.

- Boff, K. R., & Lincoln, J. E. (1988). *Engineering data compendium: Human perception and performance*. Ohio: Wright-Patterson AFB, Harry G. Armstrong Aerospace Medical Research Laboratory. Retrieved from Dayton.
- Bolchini, D., Pulido, D., & Faiola, A. (2009). "Paper in screen" prototyping: an agile technique to anticipate the mobile experience. *interactions*, 16(4), 29–33.
- Boling, E., & Smith, K. M. (2012). The design case: rigorous design knowledge for design practice. *interactions*, 19(5), 48–53.
- Borchers, J. (2001). *A pattern approach to interaction design*. New York, NY: Wiley.
- Borman, L., & Janda, A. (1986). The CHI conferences: a bibliographic history. *SIGCHI Bulletin*, 17(3), 51.
- Bradley, M. M., & Lang, P. J. (1994). Measuring emotion: the self-assessment manikin and the semantic differential. *Journal of Behavior Therapy and Experimental Psychiatry*, 25(1), 49–59.
- Branscomb, L. M. (1981). The human side of computers. *IBM Systems Journal*, 20(2), 120–121.
- Brooke, J. (1996). SUS: a quick and dirty usability scale. In: P. W. Jordan, B. Thomas, B. A. Weerdmeester, & I. L. McClelland (Eds.), *Usability evaluation in industry* (pp. 189–194). London, UK: Taylor & Francis.
- Brown, C. M. (1988). *Human-computer interface design guidelines*. Norwood, NJ: Ablex Publishing.
- Brown, L. (1993). Human-computer interaction and standardization. *Standard View*, 1(1), 3–8. <https://doi.org/10.1145/174683.174684>.
- Brown, T. (2008). Design thinking. June, *Harvard Business Review*, 84–92.
- Buchenau, M., & Suri, J. F. (2000). Experience prototyping. In: *Proceedings of the conference on designing interactive systems: Processes, practices, methods, and techniques (DIS)*.
- Buxton, B. (1986). There's more to interaction than meets the eye: some issues in manual input. In: A. D. Norman & S. W. Draper (Eds.), *User centered system design: New perspectives on human-computer interaction* (pp. 319–337). Hillsdale, NJ: Lawrence Erlbaum.
- Buxton, B. (2007a). Sketching and experience design. In: *Stanford university human-computer interaction seminar (CS 547)*. Retrieved from <http://www.youtube.com/watch?v=xxlWveKV7aE>.
- Buxton, B. (2007b). *Sketching user experiences: Getting the design right and the right design*. San Francisco, CA: Morgan Kaufmann.
- Buxton, B., Lamb, M. R., Sherman, D., & Smith, K. C. (1983). Towards a comprehensive user interface management system. *SIGGRAPH Computer Graphics*, 17(3), 35–42. <https://doi.org/10.1145/964967.801130>.
- Buxton, B., & Sniderman, R. (1980). Iteration in the design of the human-computer interface. In: *Proceedings of the 13th annual meeting of the Human Factors Association of Canada*.
- Capra, M. G. (2006). *Usability problem description and the evaluator effect in usability testing*. PhD Dissertation, Blacksburg: Virginia Tech.
- Card, S. K., Moran, T. P., & Newell, A. (1980). The keystroke-level model for user performance time with interactive systems. *Communications of the ACM*, 23(7), 396–410. <https://doi.org/10.1145/358886.358895>.
- Card, S. K., Moran, T. P., & Newell, A. (1983). *The psychology of human-computer interaction*. Hillsdale, NJ: Lawrence Erlbaum.
- Carmel, E., Whitaker, R. D., & George, J. F. (1993). PD and joint application design: a transatlantic comparison. *Communications of the ACM*, 36(6), 40–48. <https://doi.org/10.1145/153571.163265>.

- Carroll, J. M., Mack, R. L., & Kellogg, W. A. (1988). Interface metaphors and user interface design. In: M. Helander (Ed.), *Handbook of human-computer interaction* (pp. 67–85). Holland: Elsevier Science.
- Carroll, J. M., & Olson, J. R. (1987). *Mental models in human-computer interaction: research issues about what the user of software knows*. Washington, DC: National Academy Press.
- Carroll, J. M., Singley, M. K., & Rosson, M. B. (1992). Integrating theory development with design evaluation. *Behaviour & Information Technology*, 11(5), 247–255.
- Carroll, J. M., & Thomas, J. C. (1982). Metaphor and the cognitive representation of computing systems. *IEEE Transactions on Systems, Man, and Cybernetics*, 12(2), 107–116.
- Carroll, J. M., & Thomas, J. C. (1988). Fun. *SIGCHI Bulletin*, 19(3), 21–24.
- Carter, P. (2007). Liberating usability testing. *interactions*, 14(2), 18–22. <https://doi.org/10.1145/1229863.1229864>.
- Castillo, J. C., & Hartson, R. (2000). Critical Incident Data and Their Importance in Remote Usability Evaluation. In: *Proceedings of the Human Factors and Ergonomics Society annual meeting*.
- Chin, J. P., Diehl, V. A., & Norman, K. L. (1988). Development of an instrument measuring user satisfaction of the human-computer interface. In: *Proceedings of the CHI conference on human factors in computing systems, Washington, DC, May 15–19*.
- Christ, R. E. (1975). Review and analysis of color coding research for visual displays. *Human Factors*, 17(6), 542–570.
- Churchill, E. F. (2009). Ps and Qs: on trusting your socks to find each other. *interactions*, 16(2), 32–36.
- Churchill, E. F. (2010). Enticing engagement. *interactions*, 17(3), 82–87. <https://doi.org/10.1145/1744161.1744180>.
- Clement, A., & Besselaar, P. V. d. (1993). A retrospective look at PD projects. *Communications of the ACM*, 36(6), 29–37. <https://doi.org/10.1145/153571.163264>.
- Cockton, G., Lavery, D., & Woolrych, A. (2003a). Changing analysts' tunes: the surprising impact of a new instrument for usability inspection method assessment. In: *Proceedings of the international conference on human-computer interaction (HCI International)*.
- Cockton, G., & Woolrych, A. (2001). Understanding inspection methods: lessons from an assessment of heuristic evaluation. In: *Proceedings of the international conference on human-computer interaction (HCI International) and IHM 2001*.
- Cockton, G., & Woolrych, A. (2002). Sale must end: should discount methods be cleared off HCI's shelves? *interactions*, 9(5), 13–18. <https://doi.org/10.1145/566981.566990>.
- Cockton, G., Woolrych, A., Hall, L., & Hindmarch, H. (2003b). Changing analysts' tunes: the surprising impact of a new instrument for usability inspection method assessment? In: P. Johnson & P. Palanque (Eds.), *Vol. XVII. People and computers*. London: Springer-Verlag.
- Constantine, L. L. (1994a). Essentially speaking. *Software Development*, 2(11), 95–96.
- Constantine, L. L. (1994b). Interfaces for intermediates. *IEEE Software*, 11(4), 96–99.
- Constantine, L. L. (1995). Essential modeling: use cases for user interfaces. *interactions*, 2(2), 34–46. <https://doi.org/10.1145/205350.205356>.
- Constantine, L. L. (2002). Process agility and software usability: toward lightweight usage-centered design. *Information Age*, 8(2), 1–10.
- Constantine, L. L., & Lockwood, L. A. D. (1999). *Software for use: A practical guide to the models and methods of usage-centered design*. Boston, MA: Addison Wesley Longman, Inc.
- Constantine, L. L., & Lockwood, L. A. D. (2003). Card-based user and task modeling for agile usage-centered design. In: *Proceedings of the CHI conference on human factors in computing systems (tutorial)*.

- Cooper, A. (2004). *The inmates are running the asylum: Why high tech products drive us crazy and how to restore the sanity* (1st ed.). Indianapolis, IN: Sams-Pearson Education.
- Cooper, A., Reimann, R., & Dubberly, H. (2003). *About Face 2.0: The essentials of interaction design*. New York, NY: John Wiley.
- Cooper, G. (1998). *Research into cognitive load theory & instructional design at UNSW*. Retrieved from <http://paedpsych.jku.at:4711/LEHRTEXTE/Cooper98.html>.
- Cordes, R. E. (2001). Task-selection bias: a case for user-defined tasks. *International Journal of Human Computer Interaction*, 13(4), 411–419.
- Costabile, M. F., Ardito, C., & Lanzilotti, R. (2010). Enjoying cultural heritage thanks to mobile technology. *interactions*, 17(3), 30–33.
- Cox, D., & Greenberg, S. (2000). Supporting collaborative interpretation in distributed Groupware. In: *Proceedings of the ACM conference on computer supported cooperative work, Philadelphia, Pennsylvania*.
- Cross, N. (2001). Design cognition: results from protocol and other empirical studies of design activity. In: C. M. Eastman, W. M. McCracken, & W. C. Newstetter (Eds.), *Design knowing and learning: Cognition in design education* (pp. 79–103). Oxford, UK: Elsevier.
- Cross, N. (2006). *Designers ways of knowing*. London, UK: Springer.
- Dearden, A. M., & Wright, P. C. (1997). Experiences using situated and non-situated techniques for studying work in context. In: *Proceedings of the INTERACT conference on human-computer interaction*.
- del Galdo, E. M., Williges, R. C., Williges, B. H., & Wixon, D. R. (1986). An evaluation of critical incidents for software documentation design. In: *Proceedings of the Human Factors and Ergonomics Society annual meeting*.
- Demarcating User eXperience Seminar (2010). Retrieved from Schloss Dagstuhl, Germany, <http://www.dagstuhl.de/10373>.
- Department of Defense (1998). *Defense Systems Software Development (Vol. DOD-STD-2167A)*.
- Desmet, P. (2003). Measuring emotions: development and application of an instrument to measure emotional responses to products. In: M. A. Blythe, A. F. Monk, K. Overbeeke, & P. C. Wright (Eds.), *Funology: From usability to enjoyment* (pp. 111–123). Dordrecht, The Netherlands: Kluwer Academic.
- Dick, W., & Carey, L. (1978). *The systematic design of instruction*. Glenview, IL: Scott, Foresman.
- Donohue, J. (1989). *Fixing fallingwater's flaws* (pp. 99–101). Architecture.
- Dormann, C. (2003). Affective experiences in the home: measuring emotion. In: *Proceedings of the conference on home oriented informatics and telematics, the networked home of the future (HOIT)*, Irvine, CA.
- Dorst, K. (2015). *Frame innovation: Create new thinking by design*. Cambridge MA: The MIT Press.
- Dourish, P. (2001). *Where the action is: The foundations of embodied interaction*. Cambridge, MA: MIT Press.
- Draper, S. W., & Barton, S. B. (1993). Learning by exploration, and affordance bugs. In: *Proceedings of the CHI conference on human factors in computing systems (INTERCHI Adjunct)*, New York, NY.
- Dray, S., & Siegel, D. (2004). Remote possibilities? International usability testing at a distance. *interactions*, 11(2), 10–17. <https://doi.org/10.1145/971258.971264>.
- Dubberly, H. (2012). What can Steve Jobs and Jonathan Ive teach us about designing? *interactions*, 19(3), 82–85. <https://doi.org/10.1145/2168931.2168948>.
- Dubberly, H., & Evenson, S. (2011). Design as learning—or “knowledge creation”—the SECI model. *interactions*, 18(1), 75–79. <https://doi.org/10.1145/1897239.1897256>.

- Dubberly, H., & Pangaro, P. (2009). What is conversation, and how can we design for it? *interactions*, 16(4), 22–28.
- Dumas, J. S., Molich, R., & Jeffries, R. (2004). Describing usability problems: are we sending the right message? *interactions*, 11(4), 24–29. <https://doi.org/10.1145/1005261.1005274>.
- Dzida, W., Wiethoff, M., & Arnold, A. G. (1993). *ERGOGuide: The quality assurance guide to ergonomic software*. Joint internal technical report of GMD (Germany) and Delft University of Technology (The Netherlands).
- Ehn, P. (1988). *Work-Oriented Design of Computer Artifacts* (1st ed.). Stockholm, Sweden: Arbetslivcentrum.
- Ehn, P. (1990). *Work-oriented design of computer artifacts* (2nd ed.). Hillsdale, NJ: Lawrence Erlbaum.
- Ekman, P., & Friesen, W. (1975). *Unmasking the face: A guide to recognizing emotions from facial clues*. Englewood Cliffs, NJ: Prentice Hall.
- Engel, S. E., & Granda, R. E. (1975). *Guidelines for man/display interfaces (TR 00.2720)*. Poughkeepsie, NY: IBM Corporation.
- Fitts, P. M. (1954). The information capacity of the human motor system in controlling the amplitude of movement. *Journal of Experimental Psychology*, 47(6), 381–391. <https://doi.org/10.1037/h0055392>.
- Fitts, P. M., & Jones, R. E. (1947). Psychological aspects of instrument display: analysis of factors contributing to 460 “pilot error” experiences in operating aircraft controls. In: H. W. Sinaiko (Ed.), *Reprinted in selected papers on human factors in the design and use of control systems (1961)* (pp. 332–358). New York, NY: Dover.
- Flanagan, J. C. (1954). The critical incident technique. *Psychological Bulletin*, 51(4), 327–358.
- Foley, J. D., & Van Dam, A. (1982). *Fundamentals of interactive computer graphics*. Reading, MA: Addison-Wesley Longman.
- Foley, J. D., Van Dam, A., Feiner, S. K., & Hughes, J. F. (1990). *Computer graphics: Principles and practice* (2nd ed.). Boston, MA: Addison-Wesley Longman Publishing Co., Inc.
- Foley, J. D., & Wallace, V. L. (1974). The art of natural graphic man-machine conversation. *ACM Computer Graphics*, 8(3), 87.
- Forlizzi, J. (2005). Robotic products to assist the aging population. *interactions*, 12(2), 16–18.
- Forlizzi, J. (2010). All look same?: a comparison of experience design and service design. *interactions*, 17(5), 60–62. <https://doi.org/10.1145/1836216.1836232>.
- Frank, B. (2006). The science of segmentation. *interactions*, 13(3), 12–13. <https://doi.org/10.1145/1125864.1125878>.
- Frishberg, N. (2006). Prototyping with junk. *interactions*, 13(1), 21–23.
- Gajendar, U. (2012). Finding the sweet spot of design. *interactions*, 19(3), 10–11.
- Gaver, W. W. (1991). Technology affordances. In: *Proceedings of the CHI conference on human factors in computing systems, New Orleans, Louisiana*.
- Gellersen, H. (2005). Smart-Its: computers for artifacts in the physical world. *Communications of the ACM*, 48(3), 66.
- Gershman, A., & Fano, A. (2005). Examples of commercial applications of ubiquitous computing. *Communications of the ACM*, 48(3), 71.
- Gibson, J. J. (1977). The theory of affordances. In: R. Shaw & J. Bransford (Eds.), *Perceiving, acting, and knowing: Toward an ecological psychology* (pp. 67–82). Hillsdale, NJ: Lawrence Erlbaum.
- Gibson, J. J. (1979). *The ecological approach to visual perception*. Boston, MA: Houghton Mifflin.



- Giesecke, F. E., Mitchell, A., Hill, H. C., Spencer, I. L., Novak, J. T., Dygdon, J. E., et al. (2018). *Case study I: The snake light*. The Companion Website for Giesecke on the Web. Retrieved from <http://www.prenhall.com/giesecke/html/cases/snake.html>.
- Gilb, T. (1987). Design by objectives. *SIGSOFT Software Engineering Notes*, 12(2), 42–49.
- Gillham, R. (2014). *The user experience of enterprise technology*. Retrieved from <http://www.foolproof.co.uk/thinking/the-user-experience-of-enterprise-technology/>.
- Gladwell, M. (2007). *Blink: The power of thinking without thinking*. New York, NY: Little, Brown and Company.
- Good, M. D. (1989). Seven experiences with contextual field research. *SIGCHI Bulletin*, 20(4), 25–32. <https://doi.org/10.1145/67243.67246>.
- Good, M. D., Spine, T., Whiteside, J. A., & George, P. (1986). User derived impact analysis as a tool for usability engineering. In: *Proceedings of the CHI conference on human factors in computing systems, New York, NY, April 13–17*.
- Good, M. D., Whiteside, J. A., Wixon, D. R., & Jones, S. J. (1984). Building a user-derived interface. *Communications of the ACM*, 27(10), 1032–1043. <https://doi.org/10.1145/358274.358284>.
- Gothelf, J., & Seiden, J. (2016). *Lean UX: Designing great products with agile teams* (2nd ed.). Sebastopol, CA: O'Reilly Media.
- Graves, M. (2012). Architecture and the Lost Art of Drawing. *The New York Times*, Retrieved from <http://www.nytimes.com/2012/09/02/opinion/sunday/architecture-and-the-lost-art-of-drawing.html>.
- Gray, W. D., & Salzman, M. C. (1998). Damaged merchandise? A review of experiments that compare usability evaluation methods. *Human Computer Interaction*, 13(3), 203–261.
- Greenbaum, J. M., & Kyng, M. (Eds.), (1991). *Design at Work: Cooperative design of computer systems*. Hillsdale, NJ: Lawrence Erlbaum.
- Grudin, J. (1989). The case against user interface consistency. *Communications of the ACM*, 32(10), 1164–1173. <https://doi.org/10.1145/67933.67934>.
- Grudin, J. (2006). The GUI shock: computer graphics and human-computer interaction. *interactions*, 13(2), 46ff. <https://doi.org/10.1145/1116715.1116751>.
- Hackman, G., & Biers, D. (1992). Team usability testing: are two heads better than one. In: *Proceedings of the Human Factors and Ergonomics Society annual meeting*.
- Hafner, K. (2007). Inside apple stores, a certain aura enchants the faithful. *The New York Times*. December 27. Retrieved from <http://www.nytimes.com/2007/12/27/business/27apple.html?ei=5124&en=6b1c27bc8cec74b5&ex=1356584400&partner=permalink&exp=permalink&pagewanted=all>.
- Hammond, N., Gardiner, M. M., & Christie, B. (1987). The role of cognitive psychology in user-interface design. In: M. M. Gardiner & B. Christie (Eds.), *Applying cognitive psychology to user-interface design* (pp. 13–52): Wiley.
- Hamner, E., Lotter, M., Nourbakhsh, I., & Shelly, S. (2005). Case study: up close and personal from Mars. *interactions*, 12(2), 30–36.
- Hansen, W. (1971). User engineering principles for interactive systems. In: *Proceedings of the fall joint computer conference, Montvale, NJ*.
- Harrison, S., Back, M., & Tatar, D. (2006). “It’s Just a Method!”: a pedagogical experiment in interdisciplinary design. In: *Proceedings of the 6th conference on designing interactive systems, University Park, PA, USA*.
- Harrison, S., & Tatar, D. (2011). On methods. *interactions*, 18(2), 10–11. <https://doi.org/10.1145/1925820.1925823>.
- Harrison, S., Tatar, D., & Sengers, P. (2007). The three paradigms of HCI. In: *Proceedings of the Alt.chi, CHI conference on human factors in computing systems, San Jose, CA*.

- Hartson, R. (2003). Cognitive, physical, sensory, and functional affordances in interaction design. *Behaviour & Information Technology*, 22(5), 315–338.
- Hartson, R., Andre, T. S., & Williges, R. C. (2003). Criteria for evaluating usability evaluation methods. *International Journal of Human Computer Interaction*, 15(1), 145–181.
- Hartson, R., & Castillo, J. C. (1998). Remote evaluation for post-deployment usability improvement. In: *Proceedings of the Conference on Advanced Visual Interfaces (AVI)*, L'Aquila, Italy.
- Hartson, R., & Smith, E. C. (1991). Rapid prototyping in human-computer interface development. *Interacting with Computers*, 3(1), 51–91.
- Hassenzahl, M. (2012). Everything can be beautiful. *interactions*, 19(4), 60–65. <https://doi.org/10.1145/2212877.2212892>.
- Hassenzahl, M., Beu, A., & Burmester, M. (2001). Engineering joy. *IEEE Software*, 18(1), 70–76.
- Hassenzahl, M., Burmester, M., & Koller, F. (2003). AttrakDiff: Ein Fragebogen zur Messung wahrgenommener hedonischer und pragmatischer Qualität (AttrakDiff: a questionnaire for the measurement of perceived hedonic and pragmatic quality). In: *Proceedings of the Mensch & Computer 2003: Interaktion in Bewegung*, Stuttgart.
- Hassenzahl, M., Platz, A., Burmester, M., & Lehner, K. (2000). Hedonic and ergonomic quality aspects determine a software's appeal. In: *Proceedings of the CHI conference on human factors in computing systems*, The Hague, The Netherlands.
- Hassenzahl, M., & Roto, V. (2007). Being and doing: a perspective on user experience and its measurement. *Interfaces*, 72, 10–12.
- Hassenzahl, M., Schöbel, M., & Trautmann, T. (2008). How motivational orientation influences the evaluation and choice of hedonic and pragmatic interactive products: the role of regulatory focus. *Interacting with Computers*, 20, 473–479.
- Hazzan, O., & Kramer, J. (2016). Assessing abstraction skills. *Communications of the ACM*, 59(12), 43–45. <https://doi.org/10.1145/2926712>.
- Heller, F., & Borchers, J. (2012). Physical prototyping of an on-outlet power-consumption display. *interactions*, 19(1), 14–17. <https://doi.org/10.1145/2065327.2065332>.
- Hertzum, M., & Jacobsen, N. E. (2003). The evaluator effect: a chilling fact about usability evaluation methods. *International Journal of Human Computer Interaction*, 15(1), 183–204.
- Hewett, T. T. (1999). Cognitive factors in design: basic phenomena in human memory and problem solving. In: *Proceedings of the CHI conference on human factors in computing systems (extended abstracts)*.
- Hinckley, K., Pausch, R., Goble, J. C., & Kassell, N. F. (1994). A survey of design issues in spatial input. In: *Proceedings of the ACM Symposium on User Interface Software and Technology*, Marina del Rey, California.
- Hix, D., & Hartson, R. (1993). *Developing user interfaces: Ensuring usability through product & process*. New York, NY: John Wiley.
- Holtzblatt, K. (1999). Introduction to special section on contextual design. *interactions*, 6(1), 30–31. <https://doi.org/10.1145/291224.291226>.
- Holtzblatt, K. (2011). What makes things cool?: intentional design for innovation. *interactions*, 18(6), 40–47. <https://doi.org/10.1145/2029976.2029988>.
- Holtzblatt, K., Wendell, J. B., & Wood, S. (2004). *Rapid contextual design: A how-to guide to key techniques for user-centered design*. San Francisco: Morgan Kaufmann.
- Honan, M. (2013). The simple complex. *Wired Magazine*, 21(2), 44.
- Hornbæk, K., & Frøkjær, E. (2005). Comparing usability problems and redesign proposals as input to practical systems development. In: *Proceedings of the CHI conference on human factors in computing systems*, Portland, Oregon, USA.



- Houben, S., Marquardt, N., Vermeulen, J., Klokmoose, C., Schöning, J., Reiterer, H., et al. (2017). Opportunities and challenges for cross-device interactions in the wild. *interactions*, 24(5), 58–63.
- Howarth, D. (2002). *Custom cupholder a shoe-in* (p. 10). Roundel, BMW Car Club Publication.
- Hudson, J. M., & Viswanadha, K. (2009). Can "wow" be a design goal? *interactions*, 16(1), 58–61. <https://doi.org/10.1145/1456202.1456217>.
- Hudson, W. (2001). How many users does it take to change a Web site? *SIGCHI Bulletin*, 6. <https://doi.org/10.1145/967222.967230>.
- Hughes, J., King, V., Rodden, T., & Andersen, H. (1994). Moving out from the control room: ethnography in system design. In: *Proceedings of the ACM conference on computer supported cooperative work, Chapel Hill, North Carolina*.
- Hughes, J., King, V., Rodden, T., & Andersen, H. (1995). The role of ethnography in interactive systems design. *interactions*, 2(2), 56–65. <https://doi.org/10.1145/205350.205358>.
- Human Factor Research Group. (1996). WAMMI Questionnaire. Retrieved from <http://www.ucc.ie/hfrg/questionnaires/wammi/index.html>.
- Human Factor Research Group. (2010). Human Factors Research Group. Retrieved from <http://www.ucc.ie/hfrg/>.
- Hutchins, E. L., Hollan, J. D., & Norman, D. A. (1986). Direct manipulation interfaces. In: D. A. Norman & S. W. Draper (Eds.), *User centered system design: New perspectives on human-computer interaction* (pp. 87–125). Hillsdale, NJ: Lawrence Erlbaum.
- Iannella, R. (1995). HyperSAM: a management tool for large user interface guideline sets. *SIGCHI Bulletin*, 27(2), 42–45. <https://doi.org/10.1145/202511.202522>.
- Isaacson, W. (2012). Keep it simple. September, *Smithsonian Magazine*, 41–49, 90–92.
- Ishii, H., & Ullmer, B. (1997). Tangible bits: towards seamless interfaces between people, bits and atoms. In: *Proceedings of the CHI conference on human factors in computing systems, Atlanta, Georgia*.
- ISO 9241-11 (1997). *Ergonomic requirements for office work with visual display terminals (VDTs) Part 11: Guidance on usability*.
- Jacob, R. J. K. (1993). Eye movement-based human-computer interaction techniques: toward non-command interfaces. In: R. Hartson & D. Hix (Eds.), Vol. 4. *Advances in human-computer interaction* (pp. 151–190). Norwood, NJ: Ablex Publishing Corporation.
- Johnson, J. (2000). Textual bloopers: an excerpt from GUI bloopers. *interactions*, 7(5), 28–48. <https://doi.org/10.1145/345242.345255>.
- Johnson, J., & Henderson, A. (2002). Conceptual models: begin by designing what to design. *interactions*, 9(1), 25–32. <https://doi.org/10.1145/503355.503366>.
- Jokela, T. (2004). When good things happen to bad products: where are the benefits of usability in the consumer appliance market? *interactions*, 11(6), 28–35. <https://doi.org/10.1145/1029036.1029050>.
- Jones, B. D., Winegarden, C. R., & Rogers, W. A. (2009). Supporting healthy aging with new technologies. *interactions*, 16(4), 48–51.
- Judge, T. K., Pyla, P. S., McCrickard, S., & Harrison, S. (2008). Affinity diagramming in multiple display environments. In: *Proceedings of the CSCW 2008 workshop on beyond the laboratory: Supporting authentic collaboration with multiple displays, San Diego, CA*.
- Kane, D. (2003). Finding a place for discount usability engineering in agile development: throwing down the gauntlet. In: *Proceedings of the conference on agile development*.

- Kantrovich, L. (2004). To innovate or not to innovate. *interactions*, 11(1), 24–31. <https://doi.org/10.1145/962342.962354>.
- Kapoor, A., Picard, R. W., & Ivanov, Y. (2004). Probabilistic combination of multiple modalities to detect interest. In: *Proceedings of the international conference on pattern recognition (ICPR)*.
- Karat, C.-M., Campbell, R., & Fiegel, T. (1992). Comparison of empirical testing and walk-through methods in user interface evaluation. In: *Proceedings of the CHI conference on human factors in computing systems, New York, NY, May 3–7*.
- Karn, K. S., Perry, T. J., & Krolczyk, M. J. (1997). Testing for power usability: a CHI 97 workshop. *SIGCHI Bulletin*, 29(4), 63–67.
- Kaur, K., Maiden, N., & Sutcliffe, A. (1999). Interacting with virtual environments: an evaluation of a model of interaction. *Interacting with Computers*, 11(4), 403–426.
- Kawakita, J. (1982). *The original KJ method*. Tokyo: Kawakita Research Institute.
- Kennedy, S. (1989). Using video in the BNR usability lab. *SIGCHI Bulletin*, 21(2), 92–95. <https://doi.org/10.1145/70609.70624>.
- Kennedy, T. C. S. (1974). The design of interactive procedures for man-machine communication. *International Journal of Man-Machine Studies*, 6, 309–334.
- Kensing, F., & Munk-Madsen, A. (1993). PD: structure in the toolbox. *Communications of the ACM*, 36(6), 78–85.
- Kern, D., & Pfleging, B. (2013). Supporting interaction through haptic feedback in automotive user interfaces. *interactions*, 20(2), 16–21.
- Kieras, D. E., & Polson, P. G. (1985). An approach to the formal analysis of user complexity. *International Journal of Man-Machine Studies*, 22, 365–394.
- Kim, J., & Moon, J. Y. (1998). Designing towards emotional usability in customer interfaces—trustworthiness of cyber-banking system interfaces. *Interacting with Computers*, 10(1), 1–29.
- Kim, J. H., Gunn, D. V., Schuh, E., Phillips, B. C., Pagulayan, R. J., & Wixon, D. (2008). Tracking real-time user experience (TRUE): a comprehensive instrumentation for complex systems. In: *Proceedings of the CHI conference on human factors in computing systems, Florence, Italy*.
- Knemeyer, D. (2015). Design thinking and UX: two sides of the same coin. *interactions*, 22(5), 66–68. <https://doi.org/10.1145/2802679>.
- Koenemann-Belliveau, J., Carroll, J. M., Rosson, M. B., & Singley, M. K. (1994). Comparative usability evaluation: critical incidents and critical threads. In: *Proceedings of the CHI conference on human factors in computing systems, Boston, Massachusetts*.
- Kolko, J. (2015a). Design thinking comes of age. September 2015, *Harvard Business Review*, 66–71.
- Kolko, J. (2015b). Moving on from requirements. *interactions*, 22(6), 22–23. <https://doi.org/10.1145/2824754>.
- Krippendorff, K. (2006). *The semantic turn: A new foundation for design*. Boca Raton, FL: CRC Press.
- Krueger, R. A., & Casey, M. A. (2008). *Focus groups: A practical guide for applied research* (4th ed.). Thousand Oaks, CA: Sage Publications.
- Kugler, L. (2015). Touching the virtual. *Communications of the ACM*, 58(8), 16–18.
- Kuniavsky, M. (2003). *Observing the user experience: A practitioner's guide to user research*. San Francisco, CA: Morgan Kaufmann.
- Kyng, M. (1994). Scandinavian design: users in product development. In: *Proceedings of the CHI conference on human factors in computing systems*.

- Lafrenière, D. (1996). CUTA: a simple, practical, low-cost approach to task analysis. *interactions*, 3(5), 35–39. <https://doi.org/10.1145/234757.234761>.
- Landauer, T. K. (1995). *The trouble with computers: Usefulness, usability, and productivity*. Cambridge, MA: The MIT Press.
- Landry, S. (2016). Instant gratification requires total control. *Wired Magazine*, 24(5), 56.
- Lantz, A., & Gulliksen, J. (2003). Editorial: design versus design: a Nordic perspective. *International Journal of Human Computer Interaction*, 15(1), 1–4.
- Lathan, C., Brisben, A., & Safos, C. (2005). CosmoBot levels the playing field for disabled children. *interactions*, 12(2), 14–16.
- Lavery, D., & Cockton, G. (1997). Representing predicted and actual usability problems. In: *Proceedings of the international workshop on representations in interactive software development*, London.
- Lavie, T., & Tractinsky, N. (2004). Assessing dimensions of perceived visual aesthetics of web sites. *International Journal of Human-Computer Studies*, 60, 269–298.
- LeCompte, M. D., & Preissle, J. (1993). *Ethnography and qualitative design in educational research* (2nd ed.). San Diego: Academic Press.
- Lewis, C. (1982). *Using the 'thinking-aloud' method in cognitive interface design*. (Research Report RC 9265). Yorktown Heights, NY: IBM Thomas J. Watson Research Center.
- Lewis, C., Polson, P. G., Wharton, C., & Rieman, J. (1990). Testing a walkthrough methodology for theory-based design of walk-up-and-use interfaces. In: *Proceedings of the CHI conference on human factors in computing systems*, Seattle, WA.
- Lewis, J. R. (1994). Sample sizes for usability studies: additional considerations. *The Journal of the Human Factors and Ergonomics Society*, 36, 368–378.
- Lewis, J. R. (1995). IBM computer usability satisfaction questionnaires: psychometric evaluation and instructions for use. *International Journal of Human Computer Interaction*, 7, 57–78.
- Lewis, J. R. (2002). Psychometric evaluation of the PSSUQ using data from five years of usability studies. *International Journal of Human Computer Interaction*, 14, 463–488.
- Lewis, R. O. (1992). *Independent verification and validation: A life cycle engineering process for quality software*. New York, NY: John Wiley & Sons, Inc.
- Lewis, S., Mateas, M., Palmiter, S., & Lynch, G. (1996). Ethnographic data for product development: a collaborative process. *interactions*, 3(6), 52–69. <https://doi.org/10.1145/242485.242505>.
- Likert, R. (1932). A technique for the measurement of attitudes. *Archives of Psychology*, 140, 55.
- Löwgren, J. (2004). Animated use sketches: as design representations. *interactions*, 11(6), 23–27.
- Lubow, A. (2009). The triumph of Frank Lloyd Wright. June, *Smithsonian Magazine*, 52–61.
- Lund, A. M. (1997). Expert ratings of usability maxims. *Ergonomics in Design*, 5(3), 15–20.
- Lund, A. M. (2001). Measuring usability with the USE questionnaire. *Usability & User Experience (the STC Usability SIG Newsletter)*, 8(2).
- Lund, A. M. (2004). *Measuring usability with the USE questionnaire*. Retrieved from [http://www.stcsig.org/usability/newsletter/0110\\_measuring\\_with\\_use.html](http://www.stcsig.org/usability/newsletter/0110_measuring_with_use.html).
- MacKenzie, I. S. (1992). Fitts' law as a research and design tool in human-computer interaction. *Human Computer Interaction*, 7, 91–139.
- Macleod, M., Bowden, R., Bevan, N., & Curson, I. (1997). The MUSiC performance measurement method. *Behaviour & Information Technology*, 16(4), 279–293.
- Mantei, M. M., & Teorey, T. J. (1988). Cost/benefit analysis for incorporating human factors in the software lifecycle. *Communications of the ACM*, 31(4), 428–439. <https://doi.org/10.1145/42404.42408>.

- Markopoulos, P., Ruyter, B. d., Privender, S., & Breemen, A. v. (2005). Case study: bringing social intelligence into home dialogue systems. *interactions*, 12(4), 37–44. <https://doi.org/10.1145/1070960.1070984>.
- Marsh, G. P. (1864). *Man and nature: Or, physical geography as modified by human action*. New York, NY: Charles Scribner.
- Martin, B., & Hanington, B. M. (2012). *Universal methods of design: 100 ways to research complex problems, develop innovative ideas, and design effective solutions*. Beverly, MA: Rockport Publishers.
- Mason, J. G. (1968). How to be of two minds. *Nation's Business*, (October), 94–97.
- Mayhew, D. J. (1999). *The usability engineering lifecycle: A practitioner's handbook for user interface design* (1st ed.). San Francisco, CA: Morgan Kaufmann.
- McCullough, M. (2004). *Digital ground: Architecture, pervasive computing, and environmental knowing*. Cambridge, MA: MIT Press.
- McGrenere, J., & Ho, W. (2000). Affordances: clarifying and evolving a concept. In: *Proceedings of the graphics interface*.
- McInerney, P., & Maurer, F. (2005). UCD in agile projects: dream team or odd couple? *interactions*, 12(6), 19–23. <https://doi.org/10.1145/1096554.1096556>.
- Meads, J. (2010). *Personal communication with Rex Hartson*. June.
- Medlock, M. C., Wixon, D., McGee, M., & Welsh, D. (2005). The rapid iterative test and evaluation method: better products in less time. In: R. G. Bias & D. J. Mayhew (Eds.), *Cost justifying usability: An update for an internet age* (pp. 489–517). San Francisco, CA: Morgan Kaufmann.
- Medlock, M. C., Wixon, D., Terrano, M., Romero, R., & Fulton, B. (2002). Using the RITE method to improve products: a definition and a case study. In: *Proceedings of the UPA international conference, Orlando, FL*.
- Memmel, T., Gundelsweiler, F., & Reiterer, H. (2007). Agile human-centered software engineering. In: *Proceedings of the British HCI Group annual conference on people and computers, University of Lancaster, United Kingdom*.
- Miller, G. A. (1956). The magical number seven, plus or minus two: some limits on our capacity for processing information. *Psychological Review*, 63(2), 81–97. <https://doi.org/10.1037/h0043158>.
- Miller, L. (2010). *Case study of customer input for a successful product*. Retrieved from [http://www.agileproductdesign.com/useful\\_papers/miller\\_customer\\_input\\_in\\_agile\\_projects.pdf](http://www.agileproductdesign.com/useful_papers/miller_customer_input_in_agile_projects.pdf).
- Miller, L., & Sy, D. (2009). Agile user experience SIG. In: *Proceedings of the CHI conference on human factors in computing systems, Boston, April 4–9*.
- Moggridge, B. (2007). *Designing interactions*. Cambridge, MA: MIT Press.
- Molich, R., Bevan, N., Butler, S., Curson, I., Kindlund, E., & Kirakowski, J. (1998). Comparative evaluation of usability tests. In: *Proceedings of the UPA international conference, Washington, DC, June*.
- Molich, R., & Dumas, J. S. (2008). Comparative usability evaluation (CUE-4). *Behaviour & Information Technology*, 27(3), 263–282.
- Molich, R., & Nielsen, J. (1990). Improving a human-computer dialogue. *Communications of the ACM*, 33(3), 338–348. <https://doi.org/10.1145/77481.77486>.
- Molich, R., Thomsen, A. D., Karyukina, B., Schmidt, L., Ede, M., van Oel, W., et al. (1999). Comparative evaluation of usability tests. In: *Proceedings of the CHI conference on human factors in computing systems (extended abstracts), Pittsburgh, Pennsylvania*.
- Moore, N. C. (2017). How to disrupt: lessons from Tony Fadell. *The Michigan Engineering, Spring, 2017*, 56–63.

- Moran, T. P. (1981a). The command language grammar: a representation for the user interface of interactive computer systems. *International Journal of Man-Machine Studies*, 15(1), 3–50.
- Moran, T. P. (1981b). Guest editor's introduction: an applied psychology of the user. *ACM Computing Surveys*, 13(1), 1–11. <https://doi.org/10.1145/356835.356836>.
- Morville, P., & Rosenfeld, L. (2006). *Information architecture for the World Wide Web* (3rd ed.). Sebastopol, CA: O'Reilly Media, Inc.
- Mosier, J. N., & Smith, S. L. (1986). Application of guidelines for designing user interface software. *Behaviour & Information Technology*, 5(1), 39–46.
- Muller, M. J. (1991). PICTIVE—an exploration in participatory design. In: *Proceedings of the CHI conference on human factors in computing systems, New Orleans, Louisiana*.
- Muller, M. J. (1992). Retrospective on a year of participatory design using the PICTIVE technique. In: *Proceedings of the CHI conference on human factors in computing systems, Monterey, California*.
- Muller, M. J. (2003). Participatory design: the third space in HCI. In: J. A. Jacko & A. Sears (Eds.), *The human-computer interaction handbook: Fundamentals, evolving technologies and emerging applications* (pp. 1051–1058). Mahwah, NJ: Lawrence Erlbaum.
- Muller, M. J., & Kuhn, S. (1993). Participatory design. *Communications of the ACM*, 36(4), 24–28.
- Muller, M. J., Matheson, L., Page, C., & Gallup, R. (1998). Participatory heuristic evaluation. *interactions*, 5(5), 13–18. <https://doi.org/10.1145/285213.285219>.
- Muller, M. J., Wildman, D. M., & White, E. A. (1993a). 'Equal opportunity' PD using PICTIVE. *Communications of the ACM*, 36(6), 64. <https://doi.org/10.1145/153571.214818>.
- Muller, M. J., Wildman, D. M., & White, E. A. (1993b). Participatory design. *Communications of the ACM*, 36(6), 26–27.
- Mumford, E. (1981). Participative systems design: structure and method. *Systems, Objectives, Solutions*, 1(1), 5–19.
- Murano, P. (2006). Why anthropomorphic user interface feedback can be effective and preferred by users. In: C.-S. Chen, J. Filipe, I. Seruca, & J. Cordeiro (Eds.), Vol. 7. *Enterprise information systems* (pp. 241–248). Dordrecht, The Netherlands: Springer.
- Murphy, R. R. (2005). Humans, robots, rubble, and research. *interactions*, 12(2), 37–39.
- Myers, B. A. (1989). User-interface tools: introduction and survey. *IEEE Software*, 6(1), 15–23.
- Myers, B. A. (1992). *State of the art in user interface software tools*. Retrieved from <http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.70.5148&rep=rep1&type=pdf>.
- Myers, B. A. (1993). State of the art in user interface software tools. In: R. Hartson & D. Hix (Eds.), Vol. 4. *Advances in human-computer interaction*. Norwood, NJ: Ablex.
- Myers, B. A. (1995). State of the art in user interface software tools. In: R. M. Baecker, J. Grudin, W. A. S. Buxton, & S. Greenberg (Eds.), *Readings in human-computer interaction: Toward the Year 2000* (pp. 323–343). San Francisco: Morgan-Kaufmann Publishers, Inc.
- Myers, B. A., Hudson, S. E., & Pausch, R. (2000). Past, present, and future of user interface software tools. *ACM Transactions on Computer-Human Interaction*, 7(1), 3–28.
- Myers, I. B., McCaulley, M. H., Quenk, N. L., & Hammer, A. L. (1998). *MBTI manual (a guide to the development and use of the Myers Briggs type indicator)* (3rd ed.). Palo Alto, CA: Consulting Psychologists Press.
- Nass, C., Steuer, J., & Tauber, E. R. (1994). In: *Computers are social actors. Paper presented at the CHI conference on human factors in computing systems, Boston, Massachusetts*.



- Nayak, N. P., Mrazek, D., & Smith, D. R. (1995). Analyzing and communicating usability data: now that you have the data what do you do? a CHI'94 workshop. *SIGCHI Bulletin*, 27(1), 22–30. <https://doi.org/10.1145/202642.202649>.
- Newman, W. M. (1968). A system for interactive graphical programming. In: *Proceedings of the spring joint computer conference, Atlantic City, New Jersey*.
- Newman, W. M. (1998). On simulation, measurement, and piecewise usability evaluation. G. M. Olson & T. P. Moran (Eds.), Vol. 13, Issue 3, *Commentary #10 on “damaged merchandise”*, *Human-Computer Interaction* (pp. 316–323).
- Nielsen, J. (1989). Usability engineering at a discount. In: *Proceedings of the international conference on human-computer interaction (HCI International)*, Boston, Massachusetts.
- Nielsen, J. (1990). Traditional dialogue design applied to modern user interfaces. *Communications of the ACM*, 33(10), 109–118. <https://doi.org/10.1145/84537.84559>.
- Nielsen, J. (1992). Finding usability problems through heuristic evaluation. In: *Proceedings of the CHI conference on human factors in computing systems, Monterey, California*.
- Nielsen, J. (1993). *Usability engineering*. Chestnut Hill, MA: Academic Press Professional.
- Nielsen, J. (1994). Enhancing the explanatory power of usability heuristics. In: *Proceedings of the CHI conference on human factors in computing systems, Boston, Massachusetts*.
- Nielsen, J., Bush, R. M., Dayton, T., Mond, N. E., Muller, M. J., & Root, R. W. (1992). Teaching experienced developers to design graphical user interfaces. In: *Proceedings of the CHI conference on human factors in computing systems, Monterey, California*.
- Nielsen, J., & Landauer, T. K. (1993). A mathematical model of the finding of usability problems. In: *Proceedings of the INTERACT conference on human-computer interaction and chi conference on human factors in computing systems (INTERCHI)*, Amsterdam, The Netherlands.
- Nielsen, J., & Molich, R. (1990). Heuristic evaluation of user interfaces. In: *Proceedings of the CHI conference on human factors in computing systems, Seattle, Washington*.
- Nilsson, P., & Ottersten, I. (1998). Interaction design: leaving the engineering perspective behind. In: L. E. Wood (Ed.), *User interface design: Bridging the gap from user requirements to design* (pp. 131–152). Boca Raton, FL: CRC Press.
- Norman, D. A. (1986). Cognitive engineering. In: D. A. Norman & S. W. Draper (Eds.), *User centered system design: New perspectives on human-computer interaction* (pp. 31–61). Hillsdale, NJ: Lawrence Erlbaum.
- Norman, D. A. (1990). *The design of everyday things*. New York, NY: Basic Books.
- Norman, D. A. (1998). *The invisible computer—Why good products can fail, the personal computer is so complex, and information appliances are the solution*. Cambridge, MA: MIT Press.
- Norman, D. A. (1999). Affordance, conventions, and design. *interactions*, 6(3), 38–43. <https://doi.org/10.1145/301153.301168>.
- Norman, D. A. (2002). Emotion and design: attractive things work better. *interactions*, 9(4), 36–42.
- Norman, D. A. (2004). *Emotional design: Why we love (or hate) everyday things* (1st ed.). New York, NY: Basic Books.
- Norman, D. A. (2005). Human-centered design considered harmful. *interactions*, 12(4), 14–19. <https://doi.org/10.1145/1070960.1070976>.
- Norman, D. A. (2006). Logic versus usage: the case for activity-centered design. *interactions*, 13(6), 45–63. <https://doi.org/10.1145/1167948.1167978>.
- Norman, D. A. (2007a). The next UI breakthrough, part 2: physicality. *interactions*, 14(4), 46–47. <https://doi.org/10.1145/1273961.1273986>.
- Norman, D. A. (2007b). Simplicity is highly overrated. *interactions*, 14(2), 40–41. <https://doi.org/10.1145/1229863.1229885>.



- Norman, D. A. (2008). Simplicity is not the answer. *interactions*, 15(5), 45–46.
- Norman, D. A. (2009). Systems thinking: a product is more than a product. *interactions*, 16(5), 52–54.
- Norman, D. A., & Draper, S. W. (1986). *User centered system design; new perspectives on human-computer interaction*. L. Erlbaum Associates Inc.
- Nowell, L., Schulman, R., & Hix, D. (2002). Graphical encoding for information visualization: an empirical study. In: *Proceedings of the IEEE symposium on information visualization (INFOVIS)*.
- O’Conner, P. T., & Kellerman, S. (2013). Write and wrong. *Smithsonian*, 43(109), 24.
- Object Management Group (2010). *UML*. Retrieved from <http://www.uml.org/>.
- Obrist, M., Velasco, C., Vi, C., Ranasinghe, N., Israr, A., Cheok, A., et al. (2016). Sensing the future of HCI: touch, taste, and smell user interfaces. *interactions*, 23(5), 40–49.
- Olsen, D. R., Jr. (1983). Automatic generation of interactive systems. *Computer Graphics*, 17(1), 53–57.
- O’Malley, C., Draper, S., & Riley, M. (1984). Constructive interaction: a method for studying human-computer-human interaction. In: *Proceedings of the INTERACT conference on human-computer interaction, London, UK, September 4–7*.
- Open Software Foundation (1990). *OSF/motif style guide: Revision 1.0*. Upper Saddle River, NJ: Prentice-Hall, Inc.
- Palen, L., & Salzman, M. (2002). Beyond the handset: designing for wireless communications usability. *ACM Transactions on Computer-Human Interaction*, 9(2), 125–151. <https://doi.org/10.1145/513665.513669>.
- Patel, N. S., & Hughes, D. E. (2012). Revolutionizing human-computer interfaces: the auditory perspective. *interactions*, 19(1), 34–37.
- Patton, J. (2002). Hitting the target: adding interaction design to agile software development. In: *Proceedings of the OOPSLA 2002 Practitioners Reports, Seattle, Washington*.
- Patton, J. (2008). *Twelve emerging best practices for adding UX work to Agile development*, 6/27/2008. Retrieved from [http://agileproductdesign.com/blog/emerging\\_best\\_agile\\_ux\\_practice.html](http://agileproductdesign.com/blog/emerging_best_agile_ux_practice.html).
- Patton, J. (2014). *User story mapping: Discover the whole story, build the right product*. Sebastopol, CA: O’Reilly Media.
- Paulk, M. C., Curtis, B., Chrissis, M. B., & Weber, C. (1993). *Capability maturity model for software, Version 1.1 (CMU/SEI-93-TR-24)*. Pittsburgh, PA: Carnegie Mellon University.
- Pering, C. (2002). Interaction design prototyping of communicator devices: towards meeting the hardware-software challenge. *interactions*, 9(6), 36–46.
- Petersen, M. G., Madsen, K. H., & Kjaer, A. (2002). The usability of everyday technology: emerging and fading opportunities. *ACM Transactions on Computer-Human Interaction*, 9(2), 74–105. <https://doi.org/10.1145/513665.513667>.
- Pew, R. N., & Rollins, A. M. (1975). *Dialog specification procedure (5129 (Revised Ed))*. Cambridge, MA: Bolt Beranek and Newman.
- Pinker, S. (2014). *The sense of style: The thinking person’s guide to writing in the 21st century*. New York, NY: Penguin Books.
- Pogue, D. (2011). *Appeal of iPad 2 is a matter of emotions*. Retrieved from [http://www.nytimes.com/2011/03/10/technology/personaltech/10pogue.html?\\_r=2&hpw](http://www.nytimes.com/2011/03/10/technology/personaltech/10pogue.html?_r=2&hpw).
- Pogue, D. (2012). Technology’s friction problem. April, *Scientific American*, 28.
- Porter, L. H. (1895). *Cycling for health and pleasure. An indispensable guide to the successful use of the wheel*. New York, NY: Dodd, Mead and Company.
- Potosnak, K. (1988). Getting the most out of design guidelines. *IEEE Software*, 5(1), 85–86.

- Pressman, R. (2009). *Software engineering: A practitioner's approach* (7th ed.). New York, NY: McGraw-Hill.
- Pyla, P. S., Tungare, M., Holman, J., & Pérez-Quinones, M. A. (2009). Continuous UIs for seamless task migration in MPUIs: Bridging task-disconnects. *Ambient, Ubiquitous and Intelligent Interaction: Lecture Notes in Computer Science*, 5612(Pt III), 77–85.
- Pyla, P. S., Tungare, M., & Pérez-Quinones, M. A. (2006). Multiple user interfaces: why consistency is not everything, and seamless task migration is key. In: *Proceedings of the CHI 06 workshop on the many faces of consistency in cross-platform design, Montréal, Québec, Canada*.
- Quesenberry, W. (2005). Usability standards: connecting practice around the world. In: *Proceedings of the IEEE International Professional Communication Conference (IPCC), 10–13 July 2005*.
- Radoll, P. (2009). Reconstructing Australian aboriginal governance by systems design. *interactions*, 16(3), 46–49.
- Reeves, B., & Nass, C. I. (1996). *The media equation: How people treat computers, television, and new media like real people and places*. Stanford, CA: CSLI Publications.
- Reisner, P. (1977). Use of psychological experimentation as an aid to development of a query language. *IEEE Transactions on Software Engineering*, SE-3(3), 218–229.
- Resmini, A., & Rosati, L. (2011). *Pervasive information architecture: Designing cross-channel user experiences*. Burlington, MA: Morgan Kaufmann.
- Rhee, Y., & Lee, J. (2009). A model of mobile community: designing user interfaces to support group interaction. *interactions*, 16(6), 46–51.
- Rice, J. F. (1991a). Display color coding: 10 rules of thumb. *IEEE Software*, 8(1), 86.
- Rice, J. F. (1991b). Ten rules for color coding. *Information Display*, 7(3), 12–14.
- Ries, E. (2011). *The lean startup: How today's entrepreneurs use continuous innovation to create radically successful businesses*. New York, NY: Crown Publishing Group of Random House.
- Rising, L., & Janoff, N. S. (2000). The scrum software development process for small teams. *IEEE Software*, 17(4), 26–32.
- Rogers, Y., & Bellotti, V. (1997). Grounding blue-sky research: how can ethnography help? *interactions*, 4(3), 58–63. <https://doi.org/10.1145/255392.255404>.
- Rosensblum, L. D. (2013). A confederacy of senses. *Scientific American*, 308(1), 72–75.
- Rosson, M. B., & Carroll, J. M. (2002). *Usability engineering: Scenario-based development of human-computer interaction*. San Francisco: Morgan Kaufman.
- Roto, V., Law, E. L.-C., Vermeeren, A. P. O. S., & Hoonhout, J. (2011). *User experience white paper: Bringing clarity to the concept of user experience*. Retrieved from <http://www.allaboutux.org/files/UX-WhitePaper.pdf>.
- Royce, W. W. (1970). Managing the development of large scale software systems. In: *Proceedings of the IEEE Western Electronic Show and Convention (WESCON) technical papers. Reprinted in proceedings of the ninth international conference on software engineering, Pittsburgh, 1989, August 25–28, (pp. 328–338)*. Los Angeles, CA: ACM Press.
- Russell, D. M., Streitz, N. A., & Winograd, T. (2005). Building disappearing computers. *Communications of the ACM*, 48(3), 42–48.
- Savio, N. (2010). Solving the world's problems through design. *interactions*, 17(3), 52–54.
- Schleicher, D., Jones, P., & Kachur, O. (2010). Bodystorming as embodied designing. *interactions*, 17(6), 47–51. <https://doi.org/10.1145/1865245.1865256>.
- Schleifer, A. (2008). *Yahoo! design pattern library*. June 4, UX Magazine.
- Scholtz, J. (2005). Have robots, need interaction with humans! *interactions*, 12(2), 12–14.

- Schrepp, M., Held, T., & Laugwitz, B. (2006). The influence of hedonic quality on the attractiveness of user interfaces of business management software. *Interacting with Computers*, 18(5), 1055–1069.
- Scriven, M. (1967). The methodology of evaluation. In: R. Tyler, R. Gagne, & M. Scriven (Eds.), *Perspectives of curriculum evaluation* (pp. 39–83). Chicago: Rand McNally.
- Sears, A. (1997). Heuristic walkthroughs: finding the problems without the noise. *International Journal of Human Computer Interaction*, 9(3), 213–234.
- Sellen, A., Eardley, R., Izadi, S., & Harper, R. (2006). The whereabouts clock: early testing of a situated awareness device. In: *Proceedings of the CHI conference on human factors in computing systems (extended abstracts)*.
- Shattuck, L. W., & Woods, D. D. (1994). The critical incident technique: 40 years later. In: *Proceedings of the Human Factors and Ergonomics Society annual meeting*.
- Shih, Y.-H., & Liu, M. (2007). The importance of emotional usability. *Journal of Educational Technology Usability*, 36(2), 203–218.
- Shneiderman, B. (1983). Direct manipulation: a step beyond programming languages. *IEEE Computer*, 16(8), 57–69.
- Shneiderman, B. (1998). *Designing the user interface: Strategies for effective human-computer interaction* (3rd ed.). Menlo Park, CA: Addison Wesley.
- Shneiderman, B., & Plaisant, C. (2005). *Designing the user interface: Strategies for effective human-computer interaction* (4th ed.). Reading, MA: Addison-Wesley.
- Sidner, C., & Lee, C. (2005). Robots as laboratory hosts. *interactions*, 12(2), 24–26.
- Siegel, D. A. (2012). The role of enticing design in usability. *interactions*, 19(4), 82–85. <https://doi.org/10.1145/2212877.2212895>.
- Simon, H. A. (1974). How big is a chunk? *Science*, 183(4124), 482–488.
- Simonsen, J., & Kensing, F. (1997). Using ethnography in contextual design. *Communications of the ACM*, 40(7), 82–88. <https://doi.org/10.1145/256175.256190>.
- Slivka, E. (2009). *Apple job offer 'unboxing' pictures posted* (p. 2). MacRumors. 10/05/2009. Retrieved from <http://www.macrumors.com/2009/10/05/apple-job-offer-unboxing-pictures-posted/>.
- Smith, D. C., Irby, C., Kimball, R., Verplank, B., & Harslem, E. (1989). Designing the star user interface (1982). In: *Perspectives on the computer revolution* (pp. 261–283). Norwood, NJ: Ablex Publishing.
- Smith, S. L., & Mosier, J. N. (1986). *Guidelines for designing user interface software (MTR-10090)*. Bedford, MA: Mitre Corporation.
- Soon, W. (2013). *Design tips from Don Norman!* Retrieved from <http://vorkspace.com/blog/index.php/hacking-don-norman/>.
- Souza, F. d., & Bevan, N. (1990). The use of guidelines in menu interface design: evaluation of a draft standard. In: *Proceedings of the INTERACT conference on human-computer interaction*.
- Spool, J., & Schroeder, W. (2001). Testing web sites: five users is nowhere near enough. In: *Proceedings of the CHI conference on human factors in computing systems (extended abstracts)*, Seattle, Washington.
- Stake, R. (2004). *Standards-based and responsive evaluation*. Thousand Oaks, CA: Sage Publications.
- Steve, J. (2000). *Apple's one-dollar-a-year man*. January 24th, 2000, Fortune.
- Stock, W. G., & Stock, M. (2015). *Handbook of information science*. Berlin, Germany: K G Saur Verlag GmbH & Co.
- Strijland, P. (1993). Human interface standards: can we do better? *Standard View*, 1(1), 26–30. <https://doi.org/10.1145/174683.174777>.

- Suchman, L. A. (1987). *Plans and situated actions: The problem of human-machine communication*. New York, NY: Cambridge University Press.
- Sutherland, I. E. (1963). *Sketchpad: A man-machine graphical communication system* (PhD Dissertation). Cambridge, MA: MIT.
- Sutherland, I. E. (1964). *Sketchpad: A man-machine graphical communication system*. Cambridge, UK: University of Cambridge.
- Sweller, J. (1988). Cognitive load during problem solving: effects on learning. *Cognitive Science*, 12, 257–285.
- Sweller, J. (1994). Cognitive load theory, learning difficulty, and instructional design. *Learning and Instruction*, 4(4), 295–312.
- Taylor, F. W. (1911). *The principles of scientific management*. New York, NY, USA and London, UK: Harper & Brothers.
- Theofanos, M., & Quesenbery, W. (2005). Towards the design of effective formative test reports. *Journal of Usability Studies*, 1(1), 27–45.
- Theofanos, M., Quesenbery, W., Snyder, C., Dayton, D., & Lewis, J. (2005). Reporting on formative testing—a UPA 2005 workshop report. In: *Proceedings of the UPA international conference, Montreal, Quebec, June 27–July 1*.
- Thomas, J. C., & Kellogg, W. A. (1989). Minimizing ecological gaps in interface design. *IEEE Software*, 6(1), 78–86.
- Thomas, P., & Macredie, R. D. (2002). Introduction to the new usability. *ACM Transactions on Computer-Human Interaction*, 9(2), 69–73. <https://doi.org/10.1145/513665.513666>.
- Thompson, C. (2010). The emotional gadget. *Wired Magazine*, 18(11), 66.
- Thompson, C., & Vandenbroucke, B. (2015). Good vibrations: tech that talks through your skin. *Wired Magazine*, 23(1), 26.
- Tidwell, J. (2011). *Designing interfaces: Patterns for effective interaction design* (2nd ed.). Sebastopol, CA: O'Reilly Media, Inc.
- Tohidi, M., Buxton, B., Baecker, R. M., & Sellen, A. (2006). User sketches: a quick, inexpensive, and effective way to elicit more reflective user feedback. In: *Proceedings of the Nordic conference on human-computer interaction, Oslo, Norway*.
- Truss, L. (2003). *Eats, shoots & leaves: The zero tolerance approach to punctuation*. United Kingdom: Profile Books.
- Tufte, E. R. (1983). *The visual display of quantitative data*. Cheshire, Connecticut: Graphics Press.
- Tufte, E. R. (1990). *Envisioning information*. Cheshire, Connecticut: Graphics Press.
- Tufte, E. R. (1997). *Visual explanations: Images and quantities, evidence and narrative*. Cheshire, Connecticut: Graphics Press.
- Tullis, T. S. (1990). High-fidelity prototyping throughout the design process. In: *Proceedings of the human factors and ergonomics society annual meeting, Santa Monica, CA*.
- Tullis, T. S., & Albert, B. (2008). *Measuring the user experience*. Burlington, MA: Morgan Kaufmann.
- Tullis, T. S., & Stetson, J. N. (2004). A comparison of questionnaires for assessing website usability. In: *Proceedings of the UPA international conference*.
- Tungare, M., Pyla, P. S., Gline, V., Bafna, P., Balli, U., Zheng, W., et al. (2006). Embodied data objects: tangible interfaces to information appliances. In: *Proceedings of the 44th ACM southeast conference (ACMSE)*.
- U.S. Department of Health and Human Services (2006). *Research-based web design & usability guidelines*. .
- Usability Net (2006). *Questionnaire resources*. Retrieved from [http://www.usabilitynet.org/tools/r\\_questionnaire.htm](http://www.usabilitynet.org/tools/r_questionnaire.htm).

- Veer, G. C. v. d., & Melguizo, M. d. C. P. (2003). Mental models. In: *The human-computer interaction handbook* (pp. 52–80). Mahwah, NJ: Lawrence Erlbaum Associates Inc.
- Vermeeren, A. P. O. S., van Kesteren, I. E. H., & Bekker, M. M. (2003). Managing the evaluator effect in user testing. In: *Proceedings of the INTERACT conference on human-computer interaction, Zurich, Switzerland*.
- Vertelney, L. (1989). Using video to prototype user interfaces. *SIGCHI Bulletin*, 21(2), 57–61.
- Virzi, R. A. (1990). Streamlining the design process: running fewer subjects. In: *Proceedings of the human factors and ergonomics society annual meeting*.
- Virzi, R. A. (1992). Refining the test phase of usability evaluation: how many subjects is enough? *The Journal of the Human Factors and Ergonomics Society*, 34(4), 457–468.
- Wasserman, A. I. (1973). The design of 'idiot-proof' interactive programs. In: *Proceedings of the national computer conference*.
- Wasserman, A. I., & Shewmake, D. T. (1982a). Rapid prototyping of interactive information systems. *ACM SIGSOFT Software Engineering Notes: Special Issue on Rapid Prototyping*, 7(5), 171–180.
- Wasserman, A. I., & Shewmake, D. T. (1982b). Rapid prototyping of interactive information systems. In: *Proceedings of the workshop on rapid prototyping, Columbia, Maryland*.
- Webster, M. (2014). Integrating color usability components into design tools. *Communications of the ACM*, 21(3), 56–61.
- Weiser, M. (1991). The computer for the 21st century. September, *Scientific American*, 265, 94–100.
- Welie, M. v., & Hallvard, T. (2000). Interaction patterns in user interfaces. In: *Proceedings of the 7th pattern languages of programs conference, Monticello, Illinois*.
- Westerman, S., Gardner, P. H., & Sutherland, E. J. (2006). *HUMAINE D9g, taxonomy of affective systems usability testing (Workpackage 9 deliverable)*. Retrieved from <https://pdfs.semanticscholar.org/5270/0d8282b807fb1dfd4fa0f7af454ee4a89447.pdf>.
- Whiteside, J. A., Bennett, J., & Holtzblatt, K. (1988). Usability engineering: our experience and evolution. In: M. Helander (Ed.), *Handbook of human-computer interaction* (pp. 791–817). Amsterdam, The Netherlands: Elsevier Science.
- Whiteside, J. A., & Wixon, D. (1985). Developmental theory as a framework for studying human-computer interaction. In: R. Hartson (Ed.), Vol. 1. *Advances in human-computer interaction* (pp. 29–48). Norwood, NJ: Ablex Publishing.
- Whiteside, J. A., & Wixon, D. (1987). Improving human-computer interaction—a quest for cognitive science. In: *Interfacing thought: Cognitive aspects of human-computer interaction* (pp. 353–365). Cambridge, MA: MIT Press.
- Wickens, C. D., & Hollands, J. G. (2000). *Engineering psychology and human performance* (3rd ed.). Upper Saddle River, NJ: Prentice-Hall Inc.
- Wildman, D. (1995). Getting the most from paired-user testing. *interactions*, 2(3), 21–27. <https://doi.org/10.1145/208666.208675>.
- Williges, R. C. (1982). Applying the human information processing approach to human/computer interactions. In: W. C. Howell & E. A. Fleishman (Eds.), Vol. 2. *Information processing and decision making* (pp. 83–119). Hillsdale, NJ: Lawrence Erlbaum.
- Williges, R. C. (1984). Evaluating human-computer software interfaces. In: *Proceedings of the international conference on occupational ergonomics, Toronto, May*.
- Wilson, C. (2011). *Perspective-based inspection (Method 10 in 100 user experience design and evaluation methods for your toolkit)*. March. Retrieved from <http://dux.typepad.com/dux/2011/03/>.



- Winchester, W. W., III (2009). Catalyzing a perfect storm: mobile phone-based HIV-prevention behavioral interventions. *interactions*, 16(6), 5–12.
- Wixon, D. (1995). Qualitative research methods in design and development. *interactions*, 2(4), 19–26. <https://doi.org/10.1145/225362.225365>.
- Wixon, D. (2003). Evaluating usability methods: why the current literature fails the practitioner. *interactions*, 10(4), 28–34. <https://doi.org/10.1145/838830.838870>.
- Wixon, D., Holtzblatt, K., & Knox, S. (1990). Contextual design: an emergent view of system design. In: *Proceedings of the CHI conference on human factors in computing systems, Seattle, Washington*.
- Wixon, D., & Ramey, J. (Eds.), (1996). *Field methods casebook for software design*. New York, NY: John Wiley.
- Wood, L. E. (Ed.), (1998). *User interface design: Bridging the gap from user requirements to design*. Boca Raton, FL: CRC Press.
- Wright, P., Lickorish, A., & Milroy, R. (1994). Remembering while mousing: the cognitive costs of mouse clicks. *SIGCHI Bulletin*, 26(1), 41–45. <https://doi.org/10.1145/181526.181534>.
- Wright, P. K. (2005). Rapid prototyping in consumer product design. *Communications of the ACM*, 48(6), 36–41.
- Ye, S. X., & Qiu, R. G. (2003). Global identification code scheme for promptly retrieving the pertinent information of a worldwide uniquely identifiable object. In: *Proceedings of the international conference on control and automation (ICCA)*.
- Young, R. M., Green, T. R. G., & Simon, T. (1989). Programmable user models for predictive evaluation of interface designs. In: *Proceedings of the CHI conference on human factors in computing systems*.
- Young, R. R. (2001). *Effective requirements practices*. Boston, MA: Addison-Wesley Professional.
- Zhang, P. (2009). Theorizing the relationship between affect and aesthetics in the ICT design and use context. In: *Proceedings of the international conference on information resources management, Dubai, United Arab Emirates*.
- Zhang, P., & Li, N. (2005). The importance of affective quality. *Communications of the ACM*, 48(9), 105–110.
- Zhang, Z., Basili, V., & Shneiderman, B. (1999). Perspective-based usability inspection: an empirical validation of efficacy. *Empirical Software Engineering*, 4(1), 43–69. <https://doi.org/10.1023/a:1009803214692>.
- Zieniewicz, M. J., Johnson, D. C., Wong, D. C., & Flatt, J. D. (2002). The evolution of army wearable computers. *IEEE Pervasive Computing*, 1(4), 30–40.